

Chapter 6, Problem 26P

Problem

Three capacitors, $C_1 = 5\ \mu\text{F}$, $C_2 = 10\ \mu\text{F}$, and $C_3 = 20\ \mu\text{F}$, are connected in parallel across a 200-V source. Determine:

- (a) the total capacitance,
- (b) the charge on each capacitor,
- (c) the total energy stored in the parallel combination.

Step-by-step solution

Step 1 of 6

Consider the circuit as shown in Figure 1.

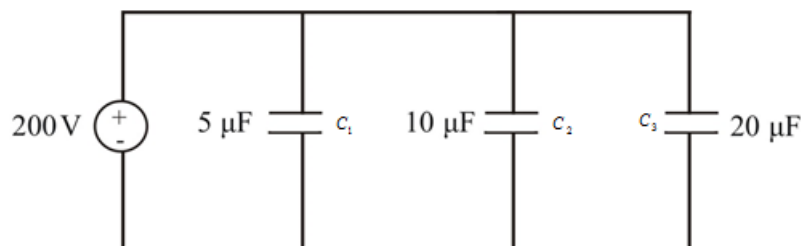


Figure 1

Step 2 of 6

(a)

Find the total capacitance C_{eq} .

The three capacitors C_1 , C_2 , and C_3 are in parallel, hence the equivalent capacitance is,

$$C_{eq} = C_1 + C_2 + C_3$$

Substitute $5\ \mu\text{F}$ for C_1 , $10\ \mu\text{F}$ for C_2 , and $20\ \mu\text{F}$ for C_3 .

$$\begin{aligned} C_{eq} &= 5\ \mu\text{F} + 10\ \mu\text{F} + 20\ \mu\text{F} \\ &= 35\ \mu\text{F} \end{aligned}$$

Thus, the total capacitance is $35\ \mu\text{F}$.

Step 3 of 6

(b)

The expression for the charge on a capacitor is,

$$q = CV$$

Since all the capacitors in parallel the voltage across all the capacitors is same.

Calculate the charge on capacitor C_1 .

$$q_1 = C_1 V$$

Substitute $5 \mu\text{F}$ for C_1 and 200 V for V .

$$\begin{aligned} q_1 &= 5 \mu \times 200 \\ &= 1 \times 10^{-3} \text{ C} \\ &= 1 \text{ mC} \end{aligned}$$

Thus, the charge on capacitor C_1 is 1 mC.

Step 4 of 6

Calculate the charge on capacitor C_2 .

$$q_2 = C_2 V$$

Substitute $10 \mu\text{F}$ for C_2 and 200 V for V .

$$\begin{aligned} q_2 &= 10 \mu \times 200 \\ &= 2 \times 10^{-3} \text{ C} \\ &= 2 \text{ mC} \end{aligned}$$

Thus, the charge on capacitor C_2 is 2 mC.

Step 5 of 6

Calculate the charge on capacitor C_3 .

$$q_3 = C_3 V$$

Substitute $10 \mu\text{F}$ for C_2 and 200 V for V .

$$\begin{aligned} q_3 &= 20 \mu \times 200 \\ &= 4 \times 10^{-3} \text{ C} \\ &= 4 \text{ mC} \end{aligned}$$

Thus, the charge on capacitor C_3 is 4 mC.

Step 6 of 6

(c)

The total energy stored in the parallel combination is,

$$W_{total} = \frac{1}{2} C_{eq} v^2$$

Substitute $35 \mu\text{F}$ for C_{eq} and 200 V for V .

$$\begin{aligned} W_{total} &= \frac{1}{2} (35 \times 10^{-6}) (200)^2 \\ &= 70 \times 10^{-2} \\ &= 0.7 \text{ J} \end{aligned}$$

Thus, the total energy stored in the parallel combination is 0.7 J.

